MICHAEL WIECK-SOSA

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EDUCATION

Carnegie Mellon University | PhD in Statistics | Advisor: Aaditya Ramdas *Sept. 2022-Present*

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by faculty to one CMU Statistics PhD student per year
- Thesis topic: high-dimensional time series | Thesis committee: Aaditya Ramdas, Cosma Shalizi, Yuting Wei, Wei Biao Wu

University of Illinois at Urbana-Champaign | MS in Statistics *May 2022*

- GPA: 3.95/4.00 | Awards: two years of guaranteed assistantships with full tuition waiver and stipend

Fordham University | BS in Mathematics with Minors in Computer Science and Economics *May 2020*

- GPA: 3.77/4.00 | Awards: *magna cum laude* | GRE: 170/170 Quantitative, 163/170 Verbal, 4.5/6.0 Writing

PROGRAMMING LANGUAGES AND TOOLS

- Python expert: extensive experience with NumPy, pandas, Polars, scikit-learn, statsmodels, and PyTorch
- R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
- C++ proficient: used for courses in Data Structures and Algorithms
- SQL, q/kdb+, Git, Bash expert: extensive experience with data management, version control, and Linux

COURSEWORK

- **Statistics**: Advanced Statistical Theory, Mathematical Statistics, Time Series Analysis, Regression, Computational Statistics
- **Computer Science**: Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
- **Math**: Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids & Lie Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

RESEARCH ASSISTANTSHIPS AND INTERNSHIPS

Carnegie Mellon University | NSF Grant | PI: Cosma Shalizi *May 2024-Present*

- Optimizing non-convex objective functions to fit complex scientific models for high-dimensional time series using Python

National Center for Supercomputing Applications | Innovative Software and Data Analysis Group *Sept. 2020-May 2022*

- Built confidence bands for concentrations & fluxes of chemicals in water over time at 1,000+ sites using R parallel computing

MIT Lincoln Lab | Group 38 *May 2021-July 2021*

- Implemented methods for tracking objects in outer space and ran simulations to compare approaches using MATLAB

University of Illinois at Urbana-Champaign | FORWARD Data Lab | Computer Science Department *Jan. 2021-May 2021*

- Modeled dynamics of misinformation spreading across multiple social media platforms using Hawkes processes with Python

Corteva Agriscience | Research & Development Division *June 2020-Aug. 2020*

- Used Gaussian mixture models to analyze changes in environments over time with 50+ variables at 1,000+ sites with Python

Fordham University | Robotics and Computer Vision Lab | Computer Science Department *May 2019-March 2020*

- Built a deep learning-based real-time tracking system for Kihansi spray toads in collaboration with zoologists using Python

TEACHING ASSISTANTSHIPS

- MS in Computational Finance program at CMU: Option Pricing, Time Series, Data Science, Machine Learning Capstone (x2)
- MS in Data Science and BS in Statistics/Machine Learning programs at CMU: Time Series, Advanced Data Analysis

PROFESSIONAL SERVICE

- Reviewer (1 paper) | Journal of the Royal Statistical Society, Series B (Statistical Methodology) *2026*
- Chair | CMU Statistics PhD student committee for organizing career events *2024-2025*